哈爾濱工業大學
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Numerical methods of global optimization

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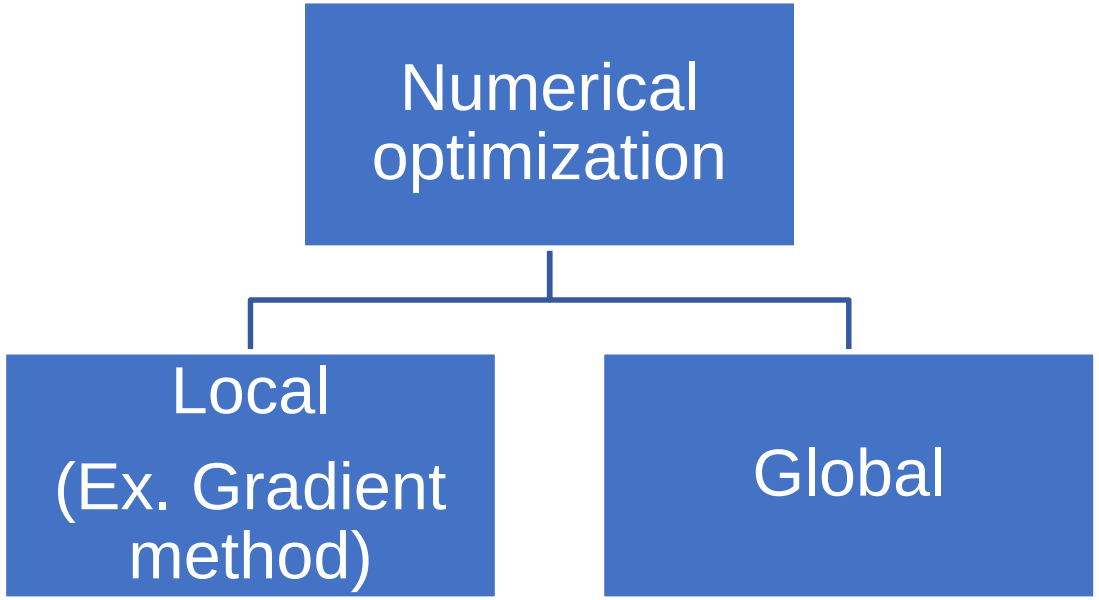
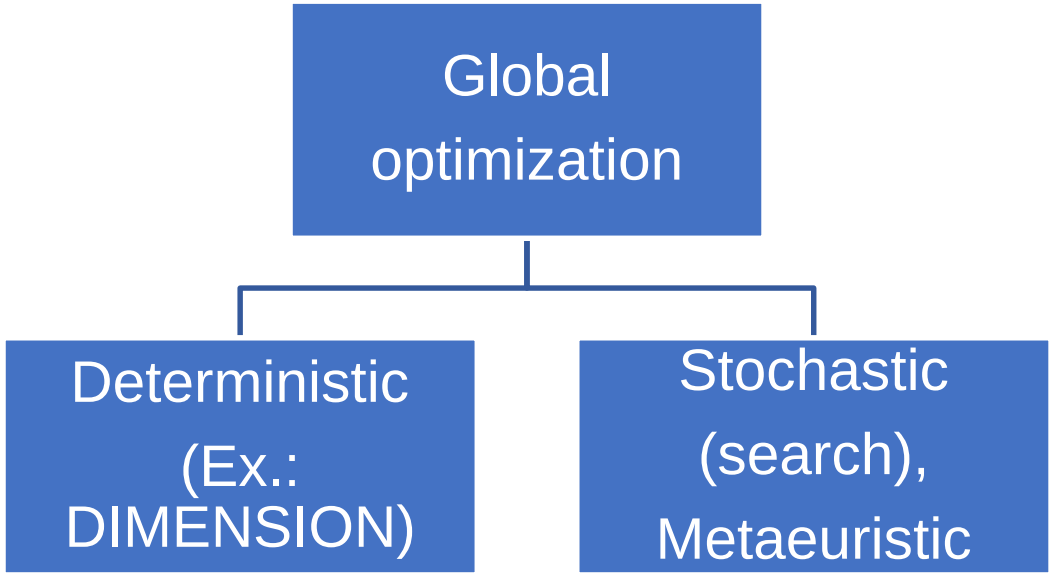
Optimization problem

Optimization problem: Let an objective function F(z) be defined in a domain Ω; find the greatest or least value of F(z) in Ω.



Global optimization methods

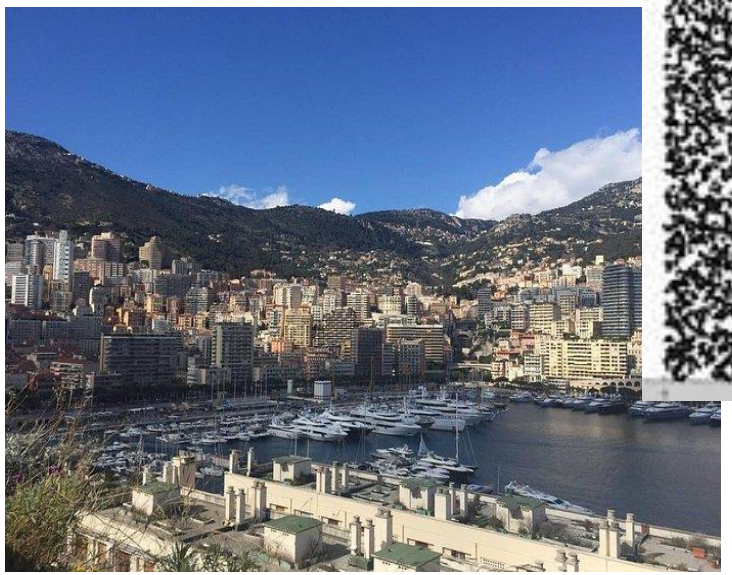


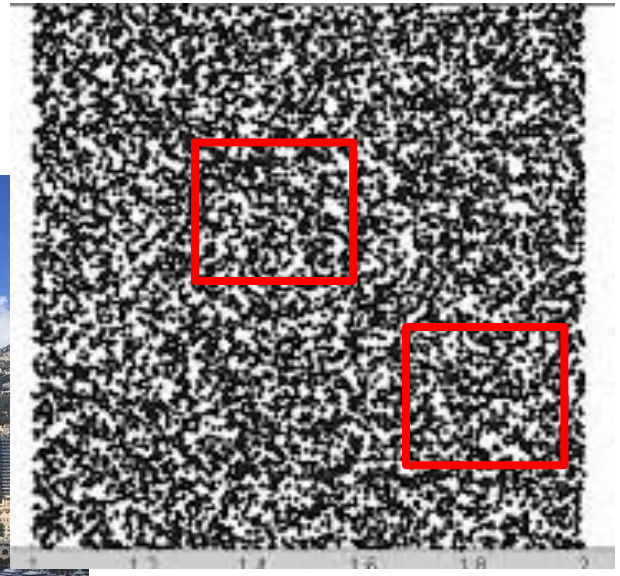
Classes of stochastic algorithms

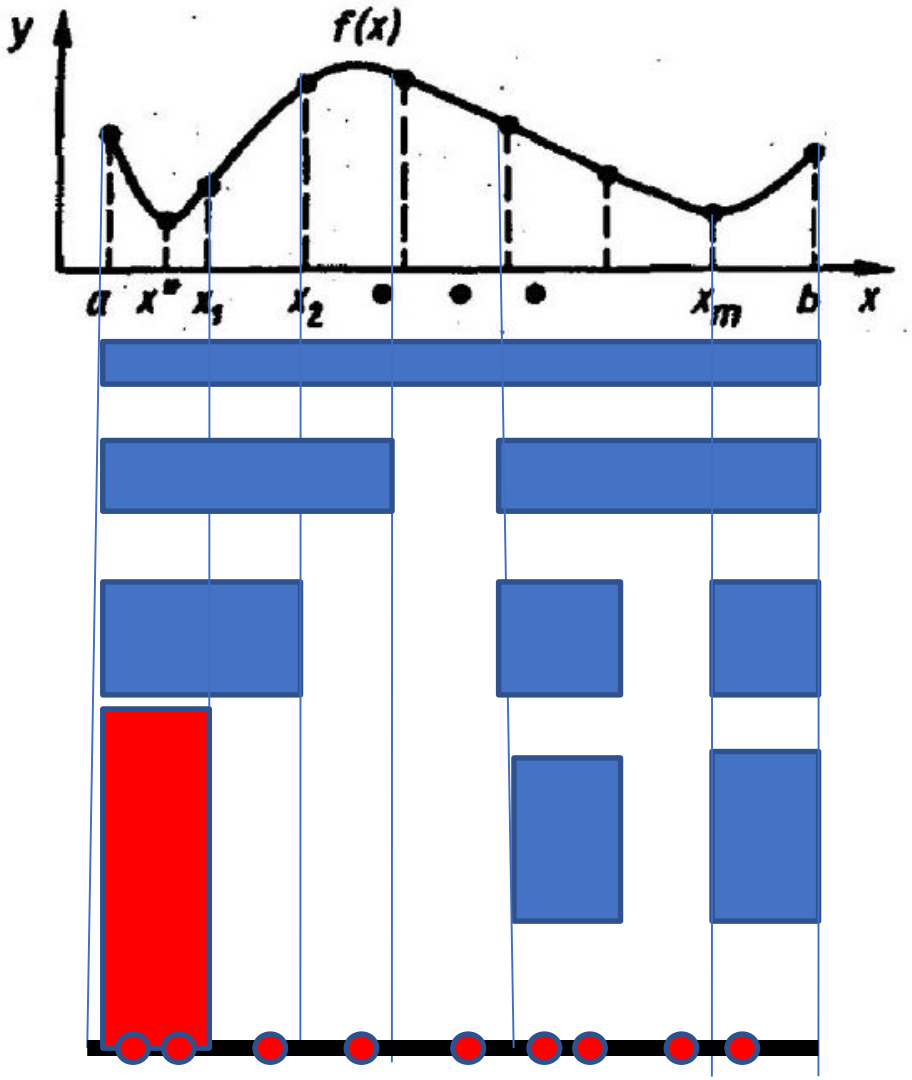
- Random search (1949, Monte Carlo method, CRS);
- Evolutionary genetic algorithms (1975, GA, 1995, DE) which are based on the ideas of biological evolution;
- Swam intelligence algorithms (PSO, 1995) based on the principles of social behavior;
- Nature inspired algorithms (annealing method) that model some physical processes and others.

Monte Carlo method

- *Everywhere dense sequence*
- $p_i \to 0$, $p_1 \to 1$







Genetic algorithm

- Setting the objective function (fitness) for individuals of the population
- Creating the initial population (Beginning of the cycle)
- Reproduction (crossover)
- Mutation
- Calculate the value of the objective function for all individuals
- Formation of a new generation (selection)
- If the stopping conditions are met, then (end of the cycle), otherwise (beginning of the cycle).
- The initial population includes N individuals (points)

Choice of Parents

- **Panmixia** - both parents are chosen randomly, each individual in the population has an equal chance of being chosen
- **Inbreeding** - the first parent is chosen randomly, and the second parent is the one that is most similar to the first parent
- **Outbreeding** - the first parent is chosen randomly, and the second parent is the one that is least similar to the first parent
- Inbreeding and outbreeding come in two forms: phenotypic and genotypic. In the case of the phenotypic form, similarity is measured depending on the value of the fitness function (the closer the values of the objective function, the more similar the individuals are), and in the case of the genotypic form, similarity is measured depending on the representation of the genotype (the fewer differences between the genotypes of individuals, the more similar the individuals are).

Crossover

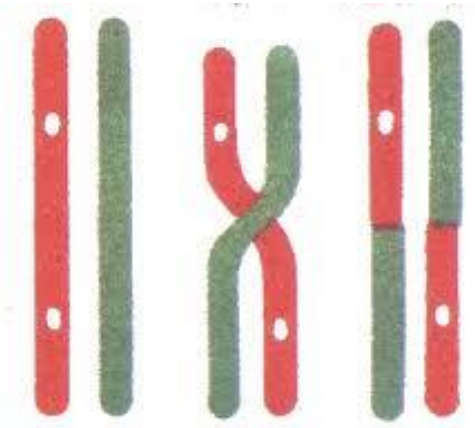
$$\boldsymbol{a}_1 \quad a_{11}, a_{12}, \ldots, a_{1m}, a_{1m+1}, \ldots, a_{1N}$$

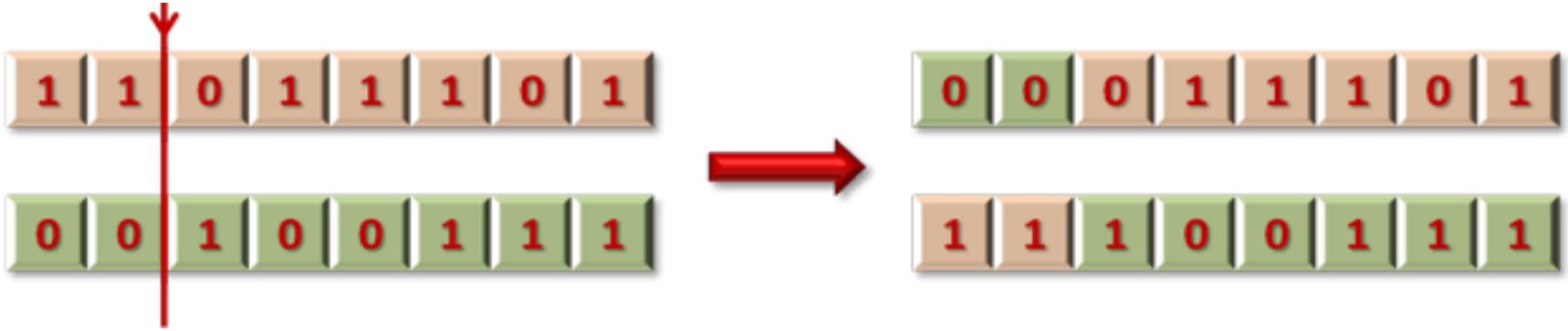
$$\boldsymbol{a}_2 \quad a_{21}, a_{22}, \ldots, a_{2m}, a_{2m+1}, \ldots, a_{2N}$$

$$\Rightarrow$$

$$\boldsymbol{a}_3 \quad a_{11}, a_{12}, \ldots, a_{1m}, a_{2m+1}, \ldots, a_{2N}$$

$$\boldsymbol{a}_4 \quad a_{21}, a_{22}, \ldots, a_{2m}, a_{1m+1}, \ldots, a_{1N}$$





Selection

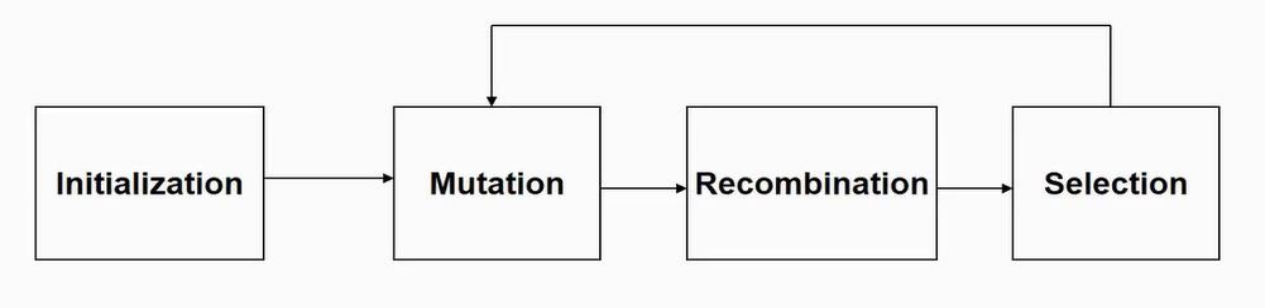
At the selection stage, it is necessary to select a certain proportion of the entire population that will remain "alive" at this stage of evolution. The remaining individuals die. The probability of an individual's survival depends on the value of the fitness function. The proportion of survivors s is a parameter of the genetic algorithm, specified in advance.

Different selection methods

- Tournament selection
- Ranking method
- Roulette method

$$p(x_i)=f(x_i)/\sum_{j=1}^{N} f(x_j)$$

Differential evolution

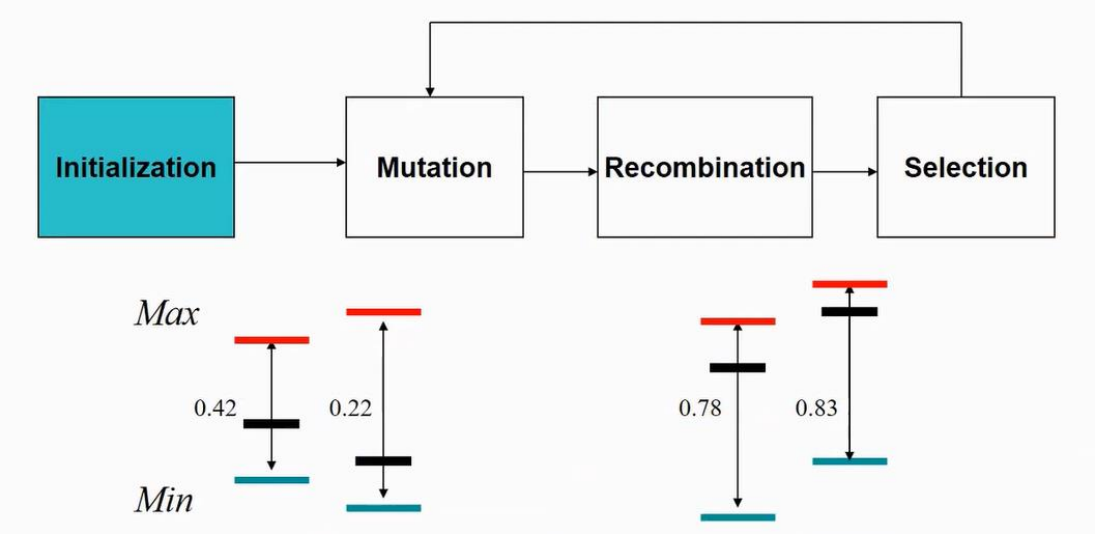


The differential evolution method was developed by Rainer Storn and Kenneth Price, first published in 1995 and further developed in their later works.

Differential evolution :

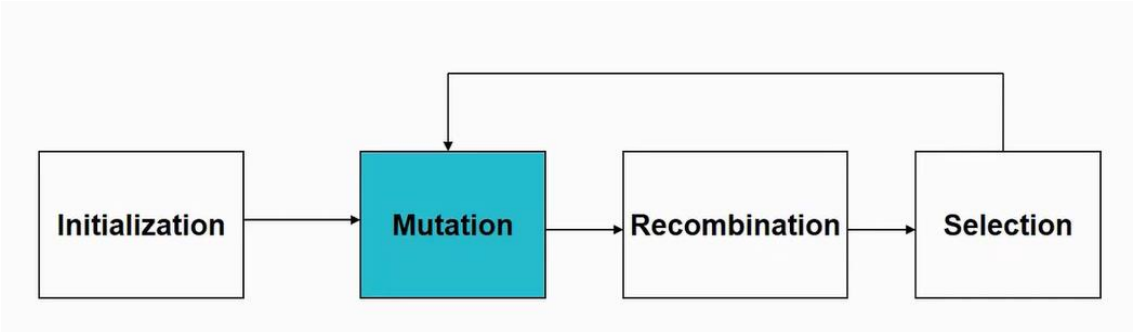
- has a small number of control parameters.
- often shows higher performance in complex optimization problems compared to other metaheuristic methods.

Differential evolution
Initialization



- Initially, a certain set of vectors, called a generation, is generated. Vectors are understood as points of an n-dimensional space in which the objective function (fitness) f(x) is defined, which must be minimized.
- At each iteration, the algorithm generates a new generation of vectors by randomly combining vectors from the previous generation. The number of vectors in each generation is the same and is one of the parameters of the method.

Differential evolution Mutation

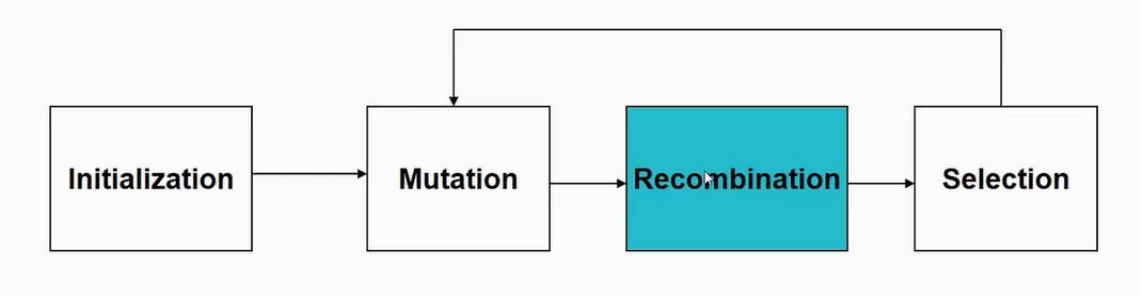


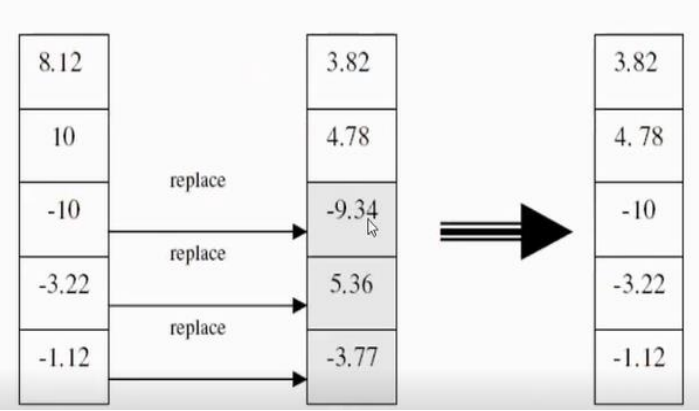
- A new generation of vectors is generated as follows. For each vector xi from the old generation, three different random vectors v1, v2, v2 are selected from among the vectors of the old generation, with the exception of the vector xi itself, and a so-called mutant vector is generated according to the formula:

$$v = v1 + F\,(v2 - v2),$$

- F is one of the parameters of the method, a certain positive real constant in the interval [0, 2].

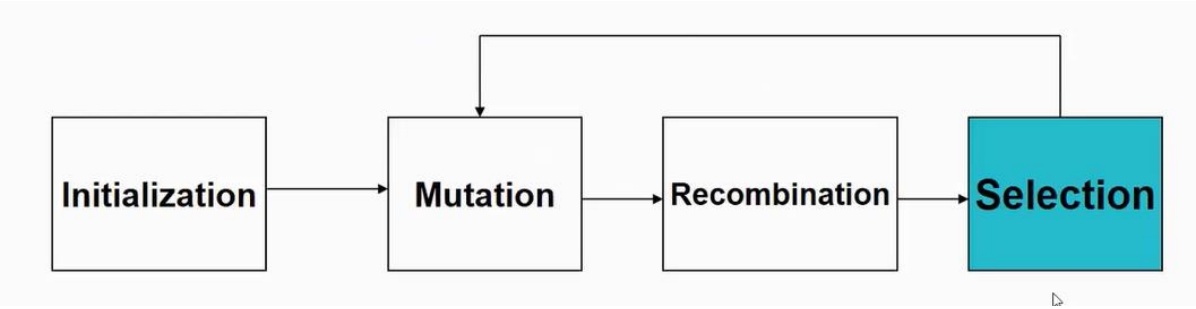
Differential evolution Recombination (Crossover)





- A “crossover” operation is performed on the mutant vector v, which consists in replacing some of its coordinates with the corresponding coordinates from the original vector xi (each coordinate is replaced with a certain probability, which is also another parameter of this method).

Differential evolution Selection



The vector obtained after crossing is called a trial vector. If it turns out to be better than the vector xi, that is,

$$f(v) < f(x_i),$$

then in the new generation the vector xi is replaced by the trial vector, and otherwise xi remains.

Drawback of metaheuristics

- The general drawback of metaheuristic algorithms is that they can’t guarantee convergence of the generated approximations to the global solution in any strict mathematical sense.
- It is impossible to guarantee that the algorithm will never fall into a false local extremum trap.
- In this regard, a situation is possible when different algorithms give different solutions to the same problem.
- Thus, the problem of developing an algorithm with justified convergence to a global optimum that is equal in performance to existing advanced evolutionary methods remains relevant

Thank you for your attention

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